

SYRIA TEL COMMUNICATION COMPANY

CUSTOMER CHURN PREDICTION

BUSINESS UNDERSTANDING

- ▶ Syria tel company is faced with customer churning to competitors and a lot of customer service calls on complains
- ▶ Customer churn leads to revenue loss and high acquisition costs
- ▶ The goal is to predict customers likely to churn early

Business Objective

- ▶ To retain valuable customers
- ▶ To increase customer satisfaction
- ▶ To reduce revenue loss
- ▶ To optimize marketing and retention efforts

Why Machine learning

- We have categorical data that has non linear regression
- Detects hidden patterns in customer behavior
- Provides early prediction of churn risk

Problem Type

- Binary Classification Problem
- 1 = Customer will churn
- 0 = Customer will stay
- Based on customer behavior features

Models Used

- Dummy Classifier (baseline model)
- Logistic Regression (linear model)
- Decision Tree (best performing model)

► Results

- Dummy: High accuracy, no real prediction power
- Logistic Regression: Good recall, low precision
- Decision Tree: Best overall performance (Accuracy, F1, AUC)
- Decision Tree selected as best model

Recommendations

- ❖ Use Decision Tree for final predictions
- ❖ Use Logistic Regression for early warning system
- ❖ Do not use Dummy model for decisions
- ❖ Focus on high-risk customers for retention

Business Actions

- Improve customer service quality through staff training
- Review pricing and competitor charges
- Offer discounts and bundle packages
- Monitor high-risk customers early

Conclusion

- ❑ Machine learning improves churn
- ❑ Decision Tree is the most effective model
- ❑ Early identification helps reduce customer loss
- ❑ Data-driven strategy improves business retention

THANK YOU
QUESTIONS



[This Photo](#) by Unknown Author is licensed under [CC BY-NC](#)